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Geothermal Potential of the Yanbu Basin in Saudi Arabia: Insights from Reservoir Modeling and Simulations

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Abstract

In line with Saudi Arabia's commitment to reach net-zero emissions by 2060, the country is actively exploring its geothermal potential, with most planned initiatives focusing on the pre-rift basement formations and geothermal provinces of the Arabian Shield. However, one of the limitations of targeting basement rocks is that they tend to have very low permeabilities, in many cases requiring to implement hydraulic fracturing to enhance connectivity. This work aims to evaluate the potential of producing geothermal energy from porous syn-rift basins' sediments through reservoir static and dynamic modeling. The methodology involves constructing a representative 3D geological model of the Al-Wajh Formation, utilizing basin and well data, along with information from analogous geological Formations. The workflow incorporates seismic data, well-logs, structural maps, and petrophysical data to create a 3D representation of the geological Formation. A methodology is introduced here to estimate porosity reduction caused by compaction for the deepest parts of the basins where temperatures tend to increase and where, in many cases, no data is available, allowing to estimate the potential production rates for these high temperature and low permeability zones. By integrating this model with numerical simulations, we provide an understanding of the potential rates and temperatures achievable in these types of basins together with the potential thermal power and electric power that could be produced. Our results indicate that water production rates of around 80 liters/second with temperatures of around 120°C can be achieved from a single production well on the basin and this rate can later be achieved on injectors where the water is re-injected into the reservoir once heat has been extracted to maintain reservoir pressure. Reservoir extension and lateral continuity are key parameters that will control well spacing and positioning. For cases where the reservoir lateral extension is large, a bigger well spacing can be used to position producers at higher depths where temperatures are higher, while injectors can be placed on shallower sections where permeability can be higher and reservoir pressure is lower. This will also allow for longer reservoir life, longer temperature break through times, and proper wastewater disposal. This study presents novel insights into the potential and effectiveness of producing geothermal energy from sedimentary rocks present on Rift Basins in Saudi Arabia. It also presents a workflow that estimates porosity reduction on the deepest parts of the Yanbu Basin based on well log data

obtained from shallower wells. The findings provide valuable guidance for potential geothermal projects in Saudi Arabia, supporting the country's climate change mitigation goals.

Introduction

Since 1950, global primary energy consumption has been drastically increasing, reflecting an increasing demand for energy worldwide. Historically, most of the energy production has been related to fossil fuels like coal, oil, and natural gas. During the last 25 years, new energy sources like solar, wind, and geothermal energy have been studied and implemented worldwide to add to the primary energy production generated from fossil fuels. By 2024, the global geothermal power generation capacity reached 16.8 GW (Cariaga, 2024). In alignment with its sustainability goals, Saudi Arabia aims to expand its geothermal capacity to 1 GW, as outlined by TAQA (2025).

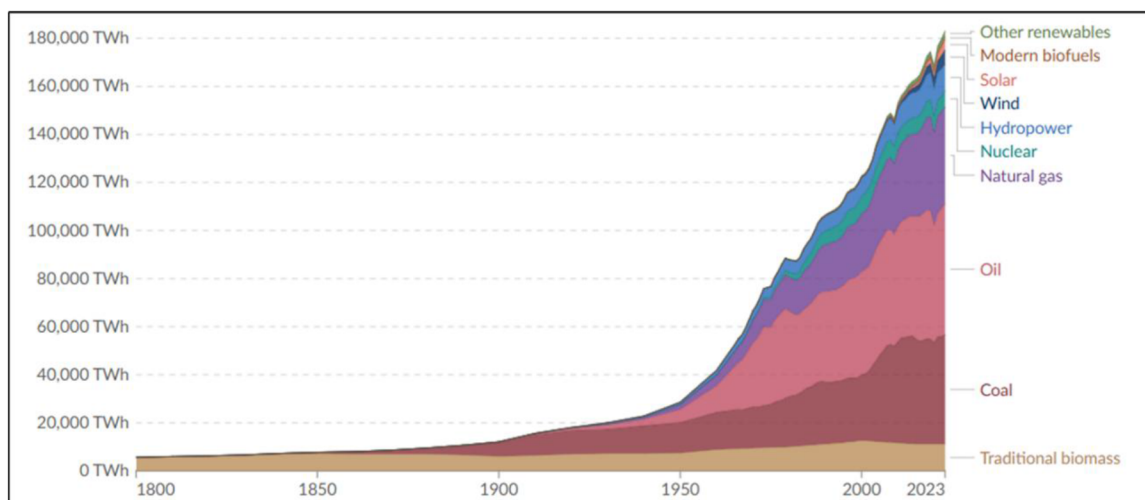


Figure 1—Global primary energy consumption by source through time. Source: Ritchie et al. (2024)

In terms of geological settings, geothermal systems can be divided into 3 main categories corresponding to hot fractured rock, hot sedimentary aquifers, and volcanic geothermal (see Fig. 2). Higher temperatures are typically related to hot fractured rock and volcanic geothermal systems. Hot sedimentary aquifers correspond to geological formations with natural porosity and permeability where the temperature is high enough for geothermal energy production. The global database of hot sedimentary aquifer geothermal projects published by Bremaud et al. (2024) showed that the average temperature of the target formation from successful projects was around 57 °C with a maximum reported temperature of 131 °C.

Aboud et al. (2021) showed that the main potential for geothermal energy production in Saudi Arabia could be located towards the western margin of the country, closer to the Red Sea Rift, where the geothermal gradient is higher (see Fig. 3). Although some geothermal projects require fracturing the rocks in subsurface to allow fluid flow, here, production associated with the primary porosity of sedimentary rocks within rift basins located on the western margin of Saudi Arabia without the need for hydraulic fracturing to enhance productivity will be evaluated.

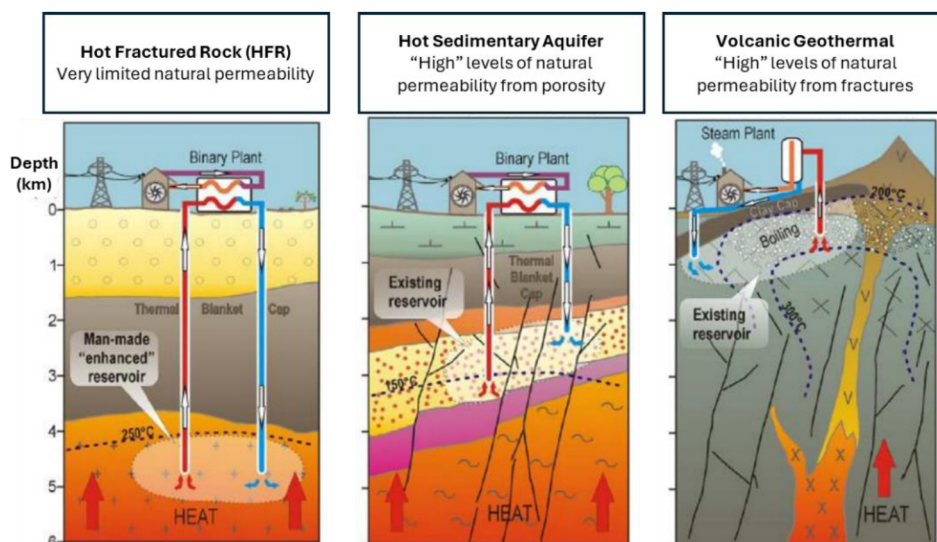


Figure 2—Geothermal Systems. Modified from Sunjay (2017)

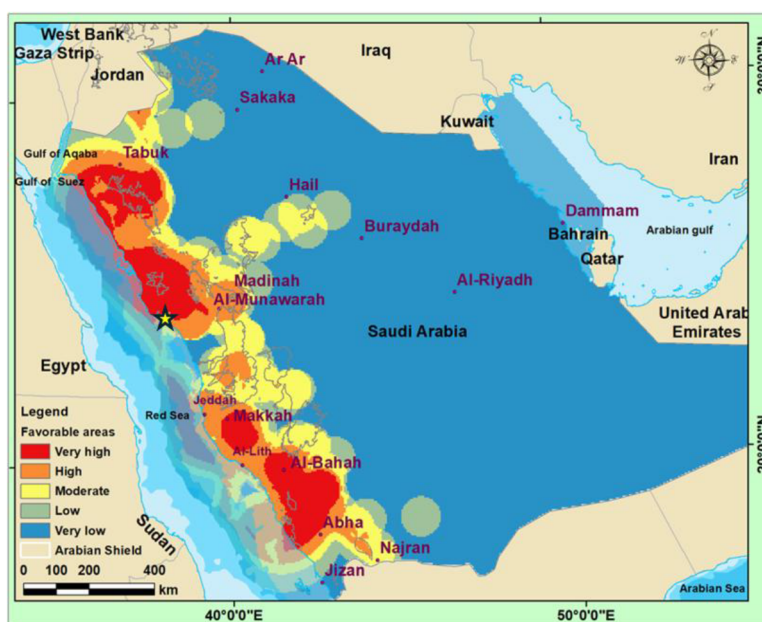


Figure 3—Geothermal Favorability Map in Saudi Arabia highlighting the location of the Yanbu Basin (study area). Source: Aboud et al. (2021)

The objective of this study is to evaluate the potential production rates and fluid temperatures that could be achieved on the Al-Wajh Formation of the Yanbu basin, corresponding to a hot sedimentary aquifer, based on reservoir static modeling and numerical simulation. The production temperatures and production rates are used here to estimate thermal power and electric power that could be obtained from the resulting rates per well. Two cases are simulated to reflect geothermal production using only water producers and their influence on reservoir pressure compared to geothermal energy production using injectors to provide pressure support for the producers, together with the influence of well spacing on pressure support and temperature breakthrough. This study provides information regarding potential temperatures and potential rates that could be reached in geological basins with similar characteristics along the western margin of Saudi Arabia.

Geological Background

The Yanbu Basin is located in the western margin of Saudi Arabia in the central Red Sea area. The rifting in the Red Sea was initiated during the Late Oligocene - Early Miocene, with main syn-rift subsidence during the Early Miocene (Bosworth et al., 2005).

Fig. 4 (a) reflects the evolution of the Red Sea rift system from Late Oligocene (31 Ma) to Middle Miocene (14 Ma) by Bosworth et al. (2005). During the early stages of the Red Sea rift, sub-basins with a half-graben geometry separated by accommodation zones were generated (Bosworth, 1994). During the Early Miocene, the Red Sea went through rapid spread, and rifting was initiated on the northern portion of the system, allowing the entrance of the sea from the Mediterranean Sea of the North and widespread marine deposition across the Red Sea (Bosworth et al., 2005). "The Tayran Group (Al Wajh, Musayr, and Yanbu Formations) includes marginal marine siliciclastics of the Al-Wajh Formation and represents the earliest rift-associated sediments deposited during the earliest Miocene" (Hughes & Johnson, 2005, p. 49).

The Al-Wajh Formation of the Tayran Group is a syn-rift unit that was deposited during the late Oligocene – early Miocene, marking the rift initiation and a deformation phase with a network of small half-grabens (Baby et al., 2024; Bosworth, 2015), as shown in Fig. 4 (b). The lower part of this syn-rift comprises heterogeneous and laterally discontinuous layers of sandstones and conglomerates throughout most of the rift system (Bosworth, 2015). Hughes & Johnson (2005) described that in some parts of the central Red Sea onshore basins, the Al-Wajh Formation is conformably overlain by the Musayr Formation while in other areas like the Yanbu Basin, it is overlain by the evaporites of the Yanbu Formation. "The Al Wajh Formation consists of red and gray-green mudstones and poorly-sorted, arkosic sandstones and conglomerate" (Hughes & Johnson, 2005, p. 73). "The environment of the Al Wajh Formation is interpreted as fluvio-lacustrine with marginal marine (possibly estuarine) pulses" (Hughes & Johnson, 2005, p. 76). The Tayran Group reflects a progression from siliciclastic to evaporitic, corresponding to a depositional regime controlled by a regional increase of marine influences (Hughes & Johnson, 2005).

The Al-Wajh Formation, referred to as Syn-rift unit 1 by Baby et al. (2024) was deposited between 28-22 Ma during the continental rift. Therefore, continental depositional environments would dominate during the deposition of this Formation. During 22-16 Ma, Marine flooding from the north through the Neothetys pathway occurred, and the Syn-rift unit 2 (Burqan Formation) was deposited in transitional conditions (Baby, et al. 2024). Well-log data and cuttings description derived from a well drilled in the basin showed that the Al-Wajh Formation in the Yanbu Basin shows a transition from continental facies at the bottom to marine influence on the top. Conglomerates at the base of the Formation are interpreted as high-energy braided fluvial systems, overlain by meandering fluvial systems' deposits. Parasequences were observed towards the upper part of the Formation where marine influence is marked by the presence of glauconite on cuttings (Lozano et al., 2024).

During the syn-rift stage (28-14 Ma) corresponding to the Late Oligocene – Early Miocene, most of the sediments that were being deposited into the Yanbu Basin as the Al-Wajh Formation came from the southeast (Baby et al., 2024). Moreover, in half-graben basins, steeper slopes near fault scarps can contribute to the development of alluvial fans, while fluvial systems would be controlled by axial flow parallel to the fault (Maitra & Bassett, 2022; Weissmann et al., 2010). Based on this, it is expected to find alluvial fans near the main fault controlling the half-graben of the Yanbu Basin while the braided and meandering fluvial systems would flow parallel to the fault as axial flow. Alluvial fans are typically coarse-grained, poorly sorted and matrix supported due to the relatively short distance of sediment transport and rapid loss of flow capacity (Blair & McPherson, 2009). Because of this, these sediments could possess lower porosity and permeability compared to braided fluvial systems sediments, where fine material is removed due to the higher energy of the stream, enhancing sorting, porosity, and permeability. Because of this, the conglomerates corresponding to braided fluvial system deposits located at the bottom of the Formation will be the target reservoir for geothermal energy production.

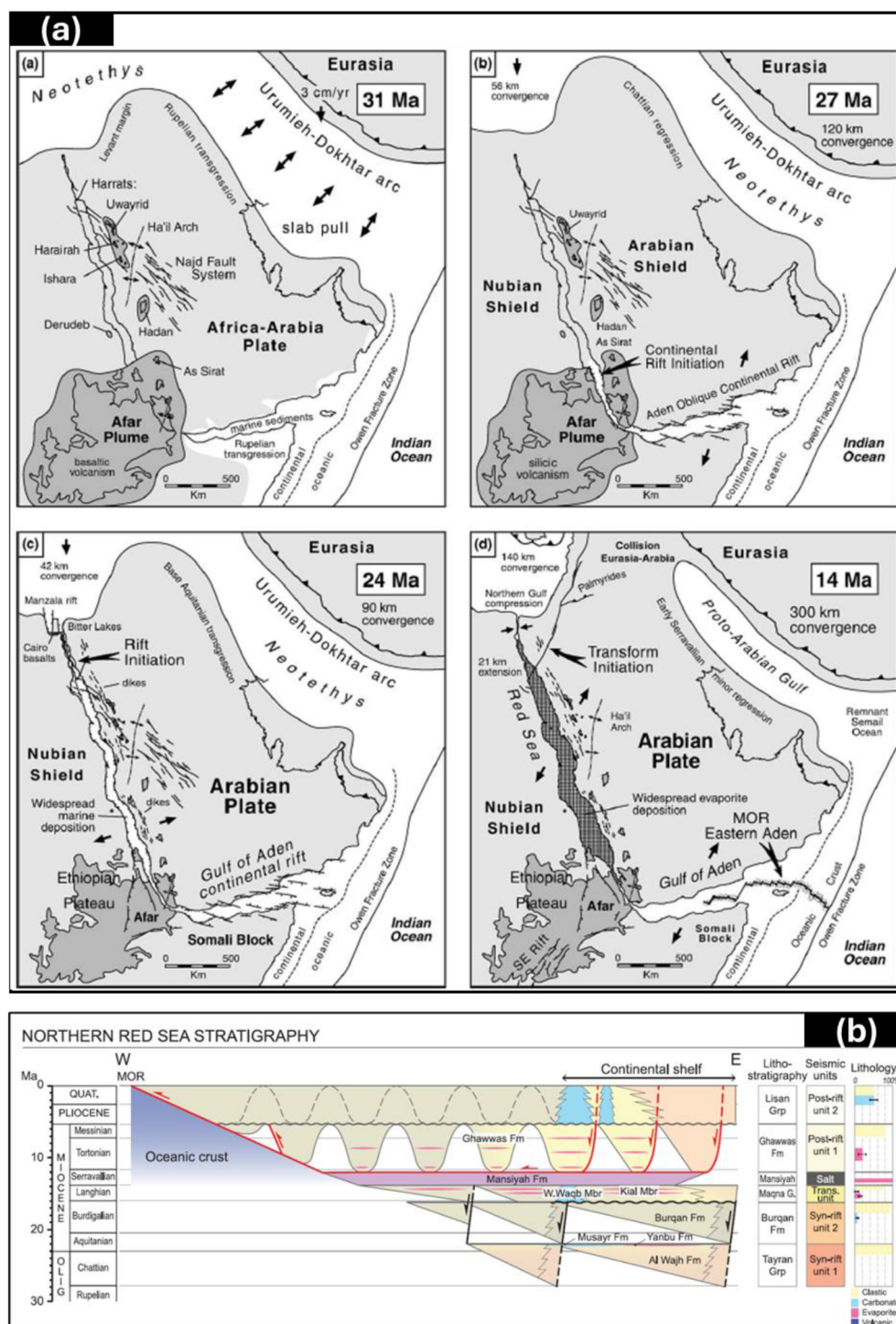


Figure 4—Tectonic evolution and stratigraphy of the Red Sea (a). Palinspatic restoration of the Red Sea – Gulf of Aden rift system. Source: [Bosworth et al. \(2005\)](#) (b). Simplified Wheeler diagrams describing the stratigraphic framework of the northern Red Sea. Source: ([Baby et al., 2024](#)).

Methodology

For this study, a static model reflecting the geology of the Al-Wajh Formation and the Yanbu Basin was generated using available data and published data. The structural model was built using seismic information together with well-log data, while the petrophysical model was built by combining well-log data for the basin and published information from analogue reservoirs. The porosity data from available wells was combined with the compaction curves shown on [Fig. 5](#) to generate a static model that properly reflected the porosity

decrease with increasing depth due to continuous burial of the sediments. Fig. 5 (a) shows a compilation of published compaction curves for sandstones done by Kim et al. (2018) which describes a low-end curve, mean curve and high-end curve for this type of rocks for multiple basins. The log data available for the basin does not reach depths beyond 2000 m and therefore, the porosity at higher depths was unknown. Therefore, the available log data for the study area was plotted together with these bounding curves, showing that the porosity vs depth trend for the Al-Wajh Formation of the Yanbu Basin falls between the low-end curve and the mean curve described on Fig. 5 (a). This observation was used to estimate the porosity in the deepest parts of the basin where no well-log data is available using the following equation:

$$\Phi_{Compaction} = \Phi_{model} \left(\frac{\Phi_{regression}}{\Phi_{reference}} \right)$$

Equation 1—Porosity correction with depth

Where Φ_{model} corresponds to the porosity derived from the log data and populated across the model, $\Phi_{regression}$ corresponds to a regression build using the low-end curve and the mean curve from Kim et al. (2018) shown on Fig. 5 (a). $\Phi_{reference}$ corresponds to the reference porosity for a given interval from the Formation derived from the regression at the well location. This way, the porosity data derived from the logs will be adjusted based on the depth of the Formation across the model.

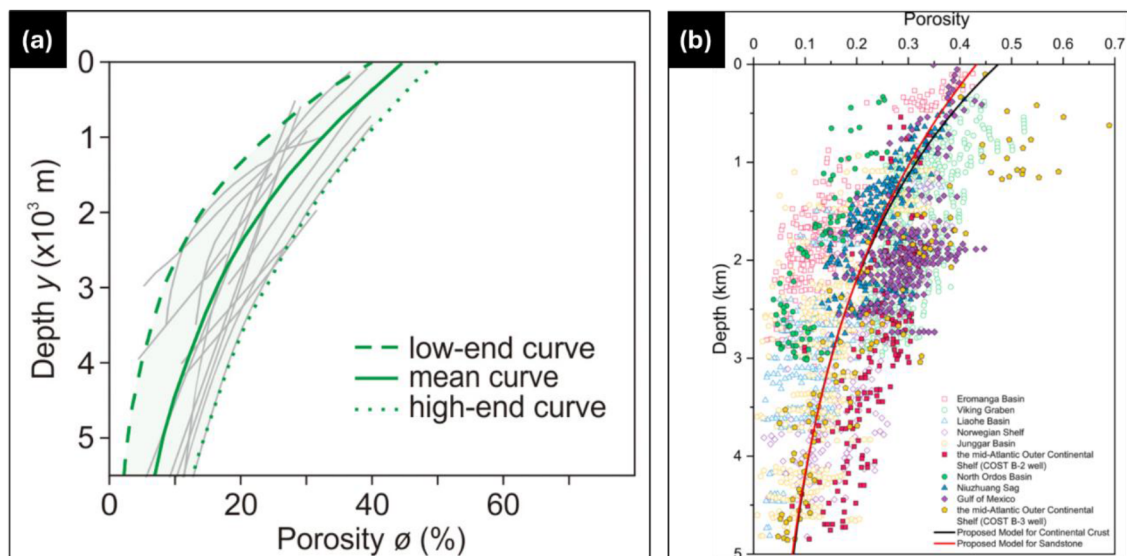


Figure 5—(a) Compilation of published compaction tendencies for sandstones by Kim et al. (2018). (b) Compilation of measured porosity data for sandstones by Chen et al. (2020)

Using the total porosity grid and a shale volume (V_{sh}) grid, effective porosity was estimated using Equation 2. Since total porosity decreases with depth, effective porosity will follow a similar trend.

$$\phi_E = \phi_T(1 - V_{sh})$$

Equation 2—Effective porosity equation. Source: Halliburton (2015).

For the permeability grid, a correlation between porosity and permeability from Elnaggar et al. (2018) for The Kareem sandstone in Egypt was used as a reference since no core data was available for the Al-Wajh Formation in the Yanbu Basin. This regression is described by Equation 3.

$$K = 1.199e^{25.39\phi}$$

Equation 3—Permeability regression for the Kareem sandstone in Egypt used for this model as analogue. Source: Elnaggar et al. (2018)

The geothermal reservoir simulations will be performed using a thermal reservoir simulator using the static model generated for the Al-Wajh Formation in the Yanbu Basin. The governing equations for the process include mass balance, energy balance and Darcy's equation corresponding to Equation 4, Equation 5, and Equation 6 (Ezekiel et al., 2024).

$$\frac{\partial(\varepsilon_{\alpha}\rho_{\alpha})}{\partial t} + \nabla \cdot (\rho_{\alpha}v_{\alpha}) = R_{\alpha}$$

Equation 4—Mass balance equation

Where ε_{α} is the volume fraction of phase α in the porous medium (porosity), ρ_{α} is the density of phase α [kg/m³], v_{α} is darcy velocity (specific discharge) of phase α [m/s], t is time [s], $\nabla \cdot (\cdot)$ is a divergence operator [1/m] and R_{α} is Source/sink term.

$$q_{\alpha} = -kA \left(\frac{k_{r\alpha}(\Delta P_{\alpha} - \rho_{\alpha}g\Delta z)}{\mu_{\alpha}} \right)$$

Equation 5—Darcy's law for multiphase flow

Where q_{α} is the volumetric flow rate of phase α through a face or cross-sectional area [m³/s], k is the absolute permeability of the porous medium [m²], A is the cross-sectional area through which the flow occurs [m²], $k_{r\alpha}$ is the relative permeability of phase α , Δp_{α} is the pressure difference of phase α across the flow path [Pa], ρ_{α} is the density of phase α [kg/m³], g is the gravitational acceleration [9.81 m/s²], Δz is the vertical distance across the flow path [m], μ_{α} is the dynamic viscosity of phase α [Pa·s].

$$\frac{\partial}{\partial t} (V\phi\rho_f U + V_s\rho_s C_s T) + \nabla \cdot (\rho_f u H) = \nabla \cdot (k_{th}\nabla T) + Q$$

Equation 6—Energy balance equation for porous media

Where V is the bulk volume of the grid block [m³], ϕ is the Porosity of the medium (fraction of bulk volume occupied by fluids), ρ_f is the fluid density [kg/m³], U is the internal energy of the fluid per unit mass [J/kg], V_s is the rock volume ρ_s is the density of the solid matrix (rock) [kg/m³], C_s is the specific heat capacity of the solid matrix [J/(kg·K)], T is the temperature [K], u is Darcy velocity of the fluid [m/s], H is enthalpy per unit mass of the fluid [J/kg], k_{th} is effective thermal conductivity of the porous medium [W/(m·K)], $\nabla \cdot (\cdot)$ is a divergence operator (net flow or flux out of a control volume) [1/m] and Q is the volumetric heat source or sink term (e.g., injection or production) [W/m³]. These equations are discretized and solved following a time-stepping scheme that progresses the solutions forward through time (Ezekiel et al., 2024; Hoteit et al., 2023; Santos et al., 2019; Z. Zhang et al., 2023).

Table 1—Simulation parameters for energy balance equation

Reservoir Thermal Conductivity	224,640 [J/m·day·C°]
Formation Water Thermal Conductivity	56,160 [J/m·day·C°]
Reservoir Heat Capacity	1'440,000 [J/m ³ ·C°]

The thermal conductivity value for the reservoir used for the simulations corresponds to the average thermal conductivity for conglomeratic rocks (J. Zhang & Zhu, 2024). Thermal conductivity of water increases with increasing temperature and decreases with salinity increase Ozbek (1980). For these simulations, water thermal conductivity will be 56,160 [J/m·day·C°] corresponding to the average thermal conductivity of water at around 100°C with a salinity of 100,000 ppm to represent the average conditions of the Al-Wajh Formation in the Yanbu Basin. The reservoir heat capacity corresponds to the average heat capacity of conglomeratic reservoirs (J. Zhang & Zhu, 2024). Water viscosity changes as a function of temperature is also represented in the simulation. Fig. 6 shows the geothermal gradient estimated using average surface temperature and maximum bottom hole temperature recordings on wells drilled in the basin corresponding to a geothermal gradient of around 37.5 °C / km. This geothermal gradient was used to build the temperature grid for the static model.

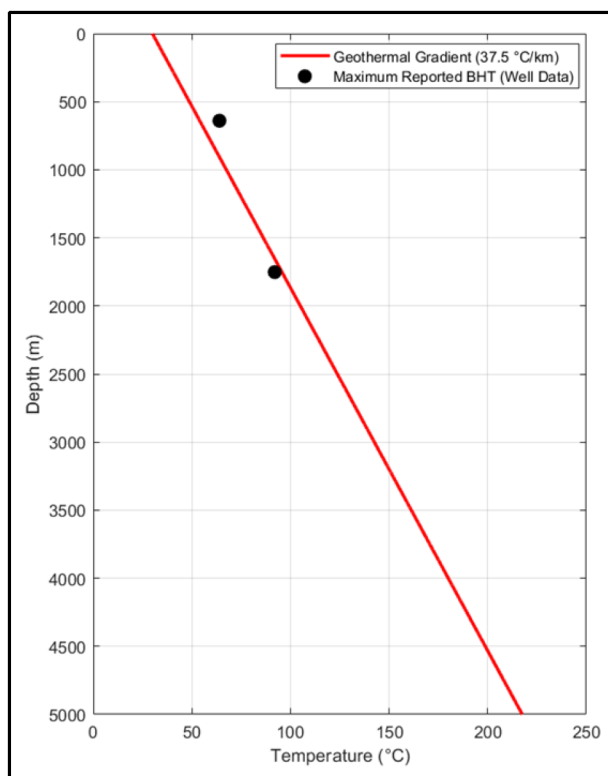


Figure 6—Expected geothermal gradient for the basin calibrated with available maximum bottom hole temperatures recorded on a well drilled in the basin. This would reflect a geothermal gradient of around 37.5°C/km

No volume modifiers will be used here on boundary conditions, since it will be assumed that some of the faults of the basin are non-conductive, making the area behave as a closed system. This will be done to assess productivity under a scenario with poor aquifer support. Two main cases will be considered during the simulations. The first case will reflect water production from 4 producers without using injector wells for pressure support. The second case will consider a scenario where the produced water is reinjected at 60 °C (considering a Carnot efficiency of 15%) through 4 injectors that will be used to provide pressure support for the 4 producers. Fig. 7 (a) shows a structural map on top of the basement on the basin and reflects the size of the sector model that will be used for the simulations, and it depicts where the production wells will be placed. The 4 producers were placed in the northwestern area of the sector model towards the deepest part of basin. Although higher depths are observed towards the southeastern region of the sector model where higher temperatures could be achieved, there would be a higher risk of finding low porosity and low permeability zones due to higher compaction (see Fig. 5 for reference) which could lead to low or null flow.

Additionally, targeting the alluvial fans can represent a higher risk since they could have lower porosity and lower permeability values due to poor sorting of the sediments and the typical matrix-supporting nature of these deposits compared to braided fluvial systems deposits (Blair & McPherson, 2009).

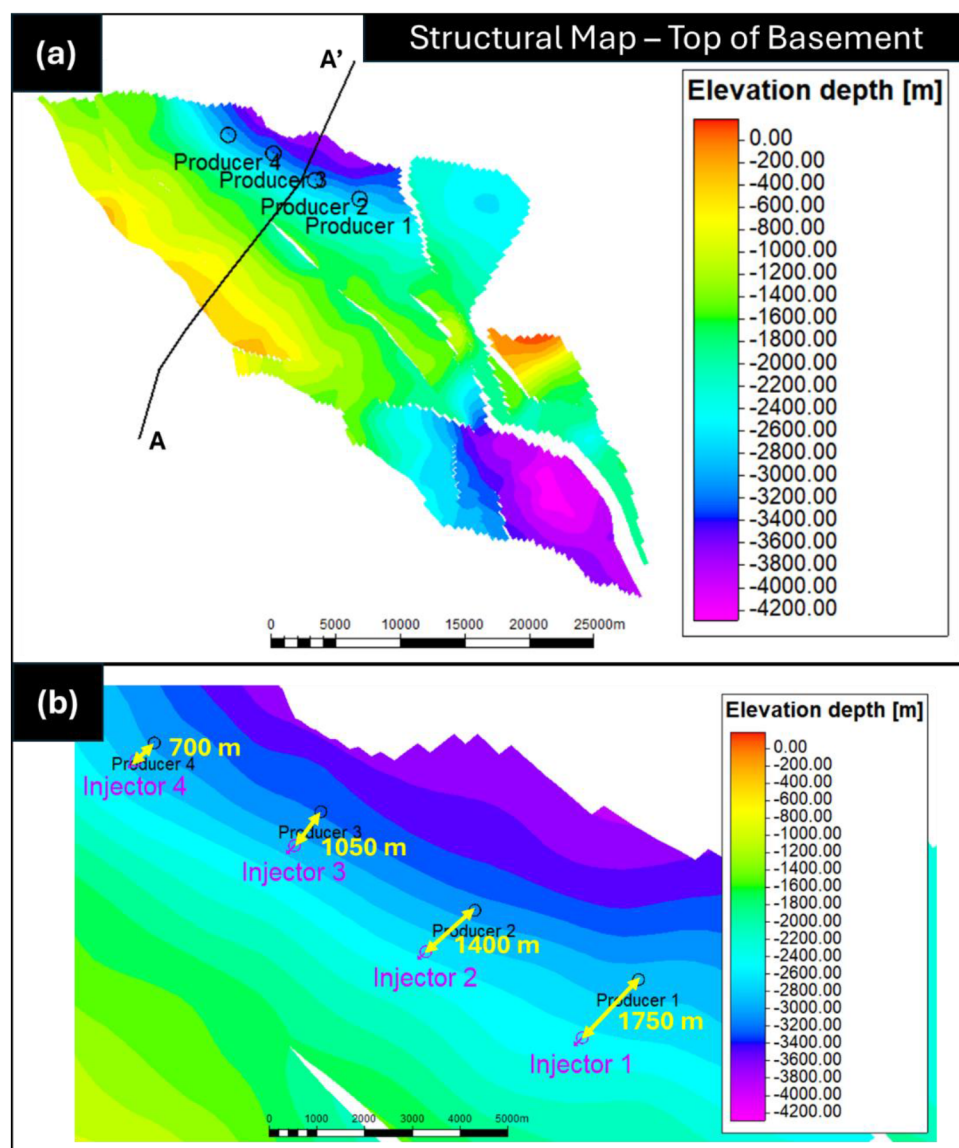


Figure 7—(a) Structural map on top of the basement highlighting the size of the sector model and the location of the 4 producers. This will be the 4 producers that will be used to simulate production without injectors. (b) Structural map on top of the basement highlighting injector and producer spacing used for the simulation of the pressure support case.

In low enthalpy geothermal projects with temperatures under 220 °C, the distance between the production zone and the re-injection zone tends to be between 0.2 km and 4 km, with an average distance of 1.28 km (Rivera Diaz et al., 2016). For the simulations, a variable well spacing distance between producer and injector of 700 m, 1050 m, 1400 m and 1750 m will be used following the ranges described by Rivera Diaz et al. (2016) as depicted on Fig. 7 (b). This will show the effect of well spacing on the thermal breakthrough, allowing the identification of an optimal well spacing for the study area.

In terms of temperature ranges, Lee (1996) described the different temperature ranges to classify geothermal projects as low-enthalpy, mid-enthalpy and high-enthalpy without a consensus. Therefore, this study will focus on estimating thermal power and electric power that could result based on production rates and water temperature. For estimating thermal power and electric power, Carnot efficiency will be

calculated based on the estimated production water temperature and the estimated rejected temperature. It will be assumed that the rejected temperature of the water will be 60 °C to maintain a Carnot efficiency between 10% and 20%, corresponding to the common range for this efficiency, using the following equation.

$$\eta = 1 - \frac{T_{out}}{T_{in}}$$

Equation 7—Carnot efficiency

Where T_{out} is the output temperature or sink temperature in Kelvin, and T_{in} is the inlet temperature or source temperature in Kelvin. Common ranges of Carnot efficiency lie between 10% and 20%. To estimate thermal power production from water production, the following equation will be used.

$$P_{thermal} = mC_{water}(T_{in} - T_{rej})$$

Equation 8—Thermal power calculation

Where m is the production mass rate in kg/day, C_{water} is the specific heat capacity of the water, which will be considered as 4.18 [kJ/kg·K] for this study. T_{in} corresponds to the initial temperature of the water, and T_{rej} corresponds to the rejected temperature, which will also be considered here as 60°C. And for estimating the electric power, the following equation will be used, which is based on Carnot efficiency and thermal power.

$$P_{electric} = \eta * P_{thermal}$$

Equation 9—Thermal power to electric power conversion

Authors like [Sanyal et al. \(2007\)](#) have described a practical upper limit for pumping rates on geothermal wells of around 160 liters/second. [Fig. 8](#) shows the production rates on global successful hot sedimentary aquifer geothermal projects targeting siliciclastic reservoirs without fractures, based on published data from [Bremaud et al. \(2024\)](#). It can be observed that the maximum production rate achieved on this type of geothermal system is about 100 liters/second, with an average of 36 liters/second.

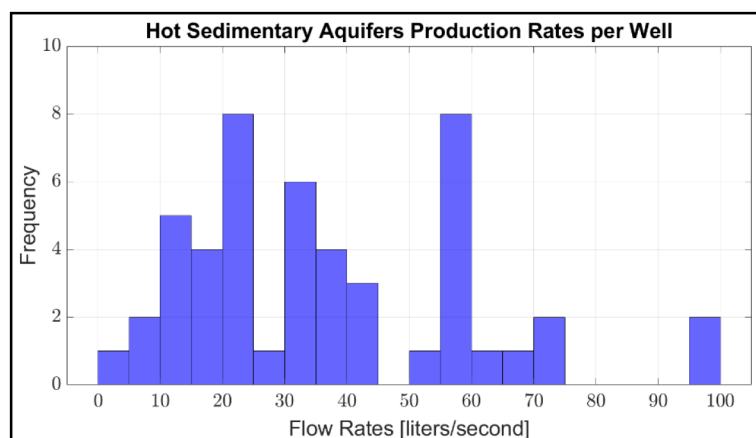


Figure 8—Water production rates on global successful hot sedimentary aquifer geothermal projects on siliciclastic reservoirs without fractures. Source of the data: [Bremaud et al. \(2024\)](#)

Results

Static Model

Fig. 9 shows the target interval for the 4 producers corresponding to the top of the conglomeratic units located at the bottom of the Al-Wajh Formation along with the expected petrophysical properties derived from the static model. In average, the production interval for the 4 producers will be located at depths ranging from 2200 m to 2600 m and the production interval of each corresponds to an interval of around 150 m. Target temperatures for these intervals will range from 112°C to 128°C. Fig. 9 (c) and Fig. 9 (d) show that the range for total porosity is between 11% and 20%, and permeability will range from 20 mD to 200 mD.

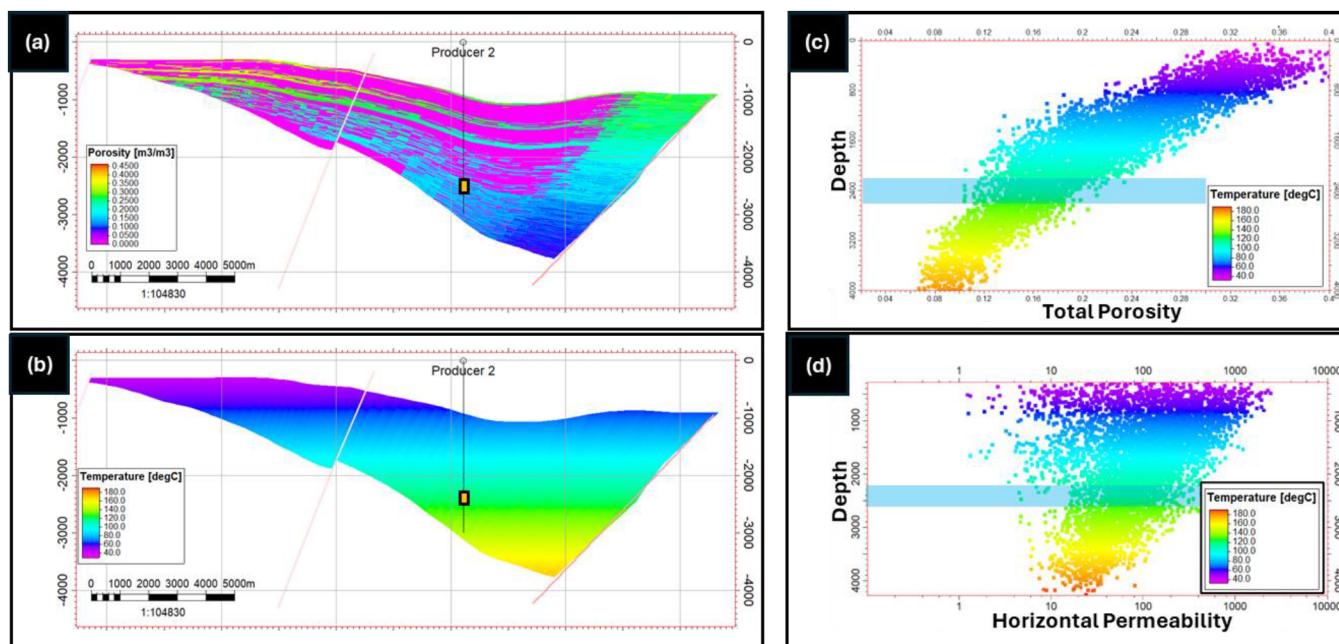


Figure 9—(a) Cross-section of the static model depicting how porosity decreases with depth. The orange polygon reflects the production interval. (b) Initial temperature grid used on the simulation. The orange polygon reflects the production interval. (c) Total porosity and temperature vs depth. The blue polygon reflects the production interval range on the producers (d) Horizontal permeability and temperature vs depth.

Fig. 10 describes the porosity and permeability ranges for multiple geothermal reservoirs worldwide corresponding to hot sedimentary aquifer geothermal projects. The area under the black dotted line represents the range for porosity and permeability where geothermal projects would require reservoir enhancement for production. The blue polygon on this figure represents the estimated ranges for porosity and permeability for the production interval for the study area as described on Fig. 9. Here, the petrophysical ranges would position the project above the dotted line, suggesting that this would correspond to a project that can be developed without requiring reservoir hydraulic fracturing for production.

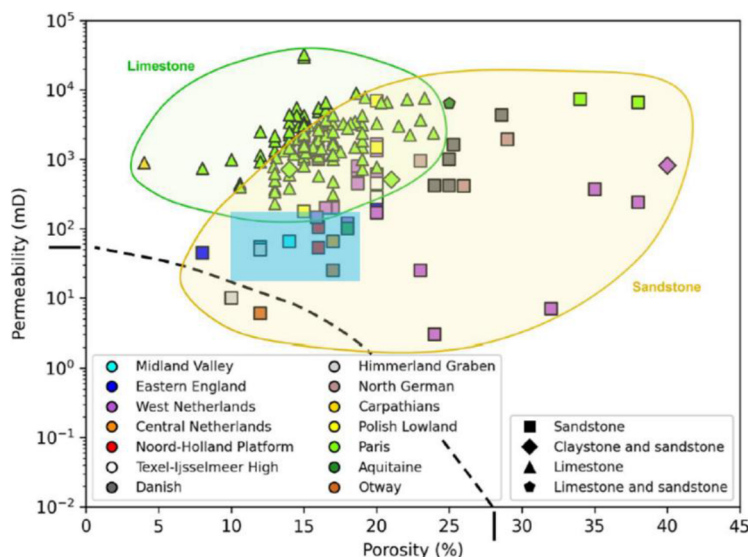


Figure 10—Porosity and permeability for multiple geothermal reservoirs worldwide. The area under the dotted black line corresponds to reservoirs that require enhanced geothermal methods for production. The blue polygon reflects the range for porosity and permeability for the production interval on the simulations. Modified from: Bremaud et al. (2024).

Simulation Results: Fluid Production Without Water Reinjection

Simulations performed using 4 producers without any volume modifiers and no injectors showed that a production rate of 80 liters/second per well could be maintained during 50 years of uninterrupted production (see Fig. 11 (a)). In mass, the production rate of 80 liters/second would correspond to a mass production of 7,430,400 kg/day, considering a water density of 1075 kg/m³ accounting for the high salinity estimated for the basin. This mass production value will be used on Equation 8 for estimating thermal power. The drawdown used during production was around 2800 kPa (406 psi). By the end of production (year 2075), reservoir pressure near the wellbore dropped to around 75% of its original pressure.

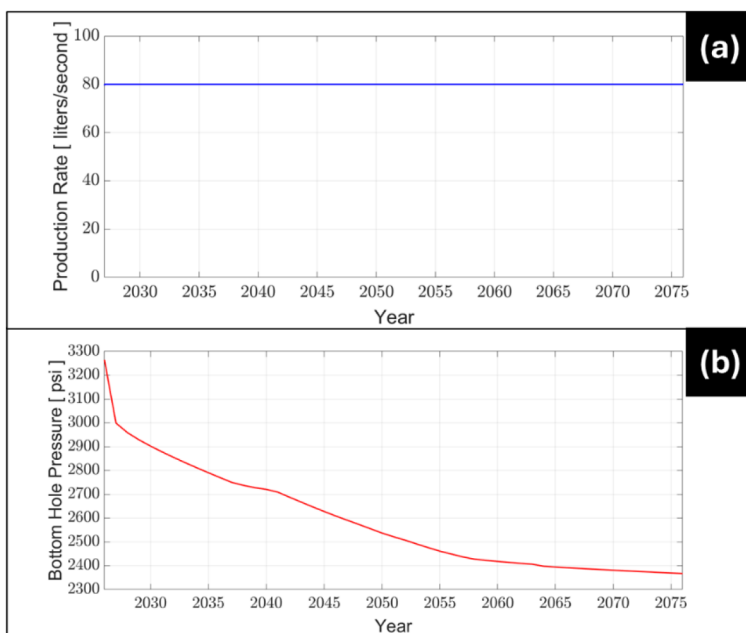


Figure 11—(a) Production rate through time for Producer 3. (b) Bottom hole pressure through time for Producer 3

Water temperature was estimated based on the completed interval with the highest permeability on the producer, considering that this would be the interval with the highest productivity on the well. Fig. 12 (a)

shows that the production temperature for most of the wells would be around 125 °C except for Producer 1 that is slightly shallower and where the interval with the highest permeability is slightly shallower, leading to lower temperatures. Fluid temperature reduction between bottom and surface due to heat transfer through the wellbore can be around 3.8% (Brown & Falcone, 2024). Based on this, a fluid temperature of 120 °C can be expected on surface conditions.

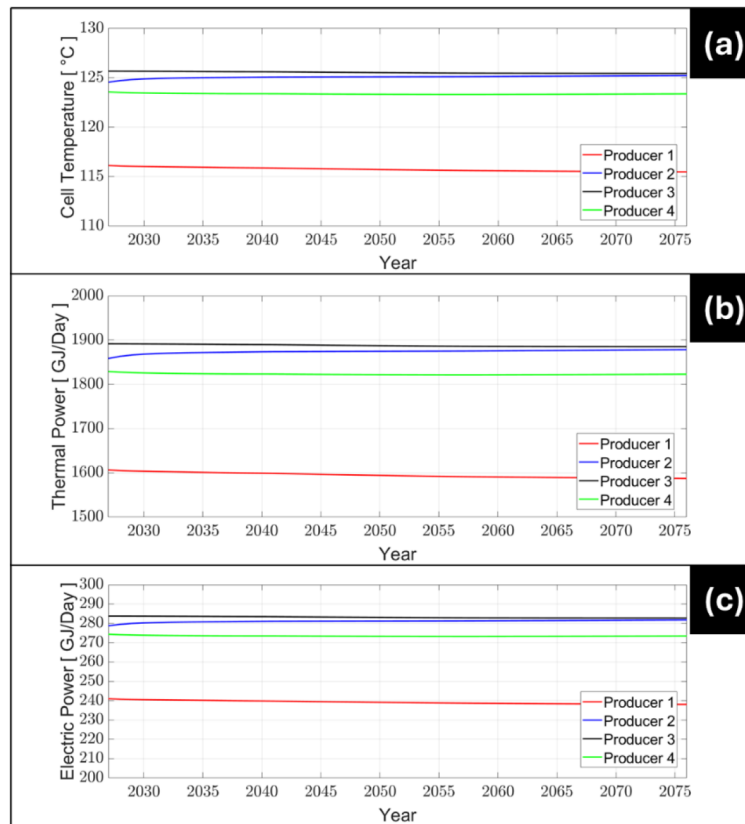


Figure 12—(a) Temperature of production cell with the highest permeability per well. (b) Thermal power generation per well through time. (c) Electric power generation assuming a Carnot efficiency of 15%

Since no injectors are being used for this case, water temperature remains almost constant during the whole process, which would result in constant thermal power and electric power as shown in Fig. 12 (a) and Fig. 12 (b). Most of the wells showed a thermal power near 1850 GJ/day (0.02 GW). It is also important to highlight here the significant drop on power when converting from thermal power to electric power. Although production rates remained constant during the 50 years of constant production, it was observed that reservoir pressure decreased across the basin.

Simulation Results: Fluid Production With Water Re-Injection For Pressure Support

For the second case, the produced water is reinjected at a temperature of 60°C (rejected temperature) starting 6 months after the water production starts. The goal of using the injectors is to both dispose of the produced water and to maintain reservoir pressure to avoid significant changes in the stress field that could result in fault reactivation and induced seismicity (Zoback, 2007). For this case, the production rate of the 4 producers is 80 liters/s and injection rate is the same considering that the amount of water that is produced is then reinjected into the reservoir. Based on Fig. 13, for Producer 3 and Producer 4, the temperature breakthrough happens faster. For these 2 producers, a temperature drop between 15 °C and 27°C is observed, bringing the temperature under 100 °C by the year 2074 on Producer 4. The temperature for Producer 2 does not decrease much in comparison, since only a reduction in temperature of around 3 °C to 6 °C is observed after

64 years of constant production and injection in both wells. In terms of pressure support, pressure drops by around 190 psi for Producer 1 and Producer 2 before receiving support from the injectors. For producers 3 and 4 pressure drops less, especially for Producer 4 where bottom hole pressure only drops 140 psi.

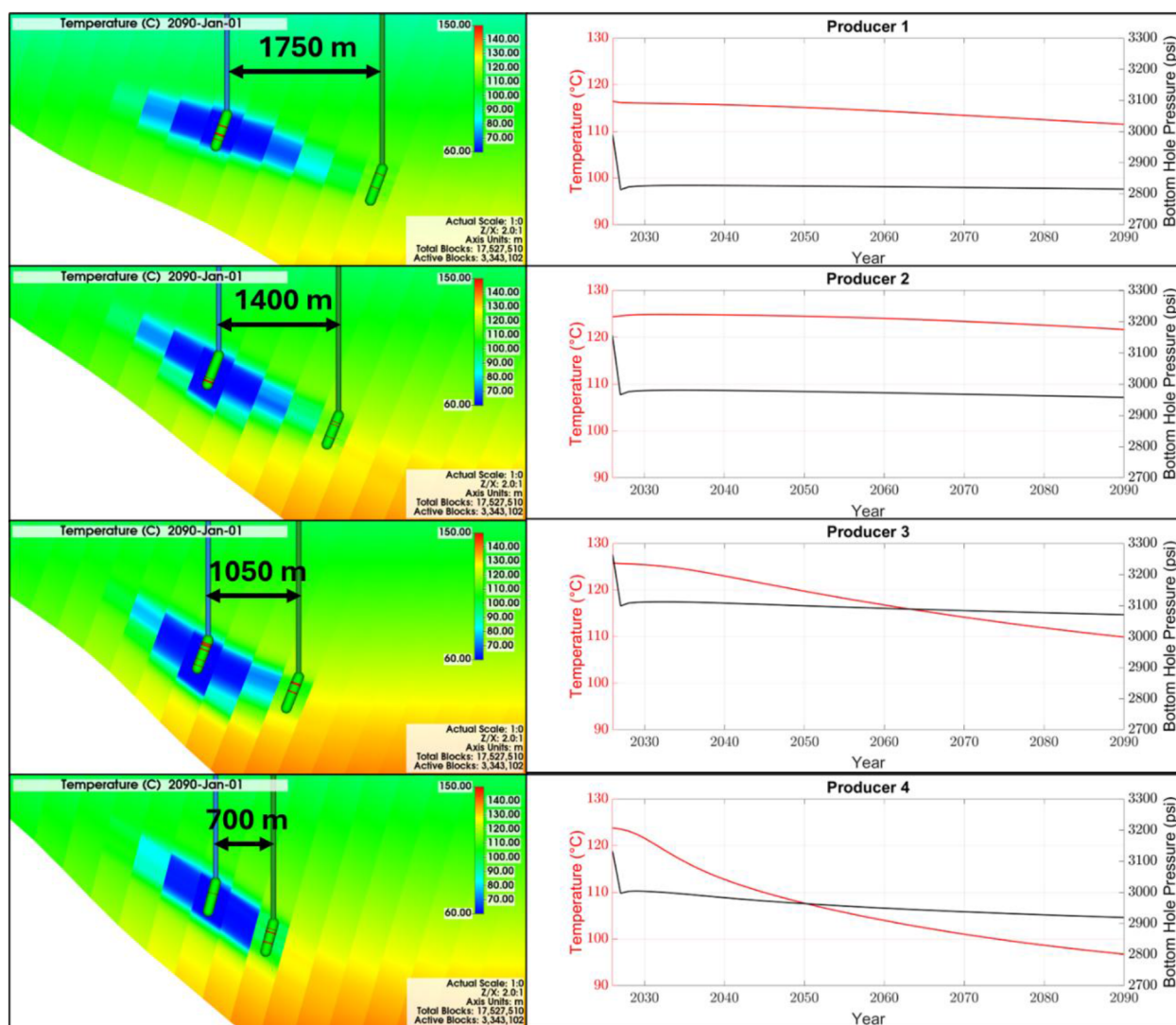


Figure 13—Thermal breakthrough and pressure support vs well spacing

For this case, since cooler water is being reinjected to maintain reservoir pressure and productivity, it is observed that produced water temperatures will not remain constant through time and will strongly depend on well spacing between producer and injectors. Therefore, thermal power and electric power production for this case is not constant and will decrease through time as temperature drops (see Fig. 14). For smaller well spacing, temperature will drop faster due to faster connectivity leading to faster drop on power production.

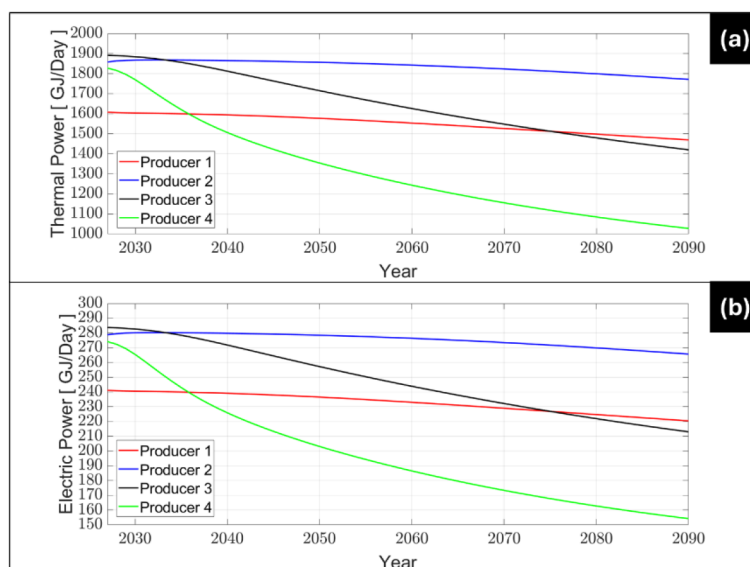


Figure 14—(a) Thermal power production through time per well. (b) Electric power production through time per well.

Discussion

It was shown that the average temperature of the produced water could be around 120 °C. Dry-steam plants and flash-steam plants require high-temperature hydrothermal sources with temperatures ranging from 149 °C to 371 °C which would not be achieved in this case. Therefore, a binary-cycle power plant would be required here where the heat from the produced water would be transferred to another fluid with a lower boiling point that will turn into steam and will drive a generator turbine (EIA, 2022). Because of this, the thermal power that could be extracted and transferred to the second fluid with lower boiling point and this process was estimated and is described by Equation 8. After this, the second fluid will be used in the binary-cycle power plant to generate electric power with an efficiency of around 15% given by the estimated Carnot efficiency (Equation 7) and Equation 9. Because of this, it can be observed that for this case, it would be better to use the thermal power directly to avoid a significant energy loss when converting thermal power to electric power.

One of the possible uses for this water is related to water desalination considering the close distance between the production site and the Yanbu City. Some desalination plants, like Multi-effect Desalination Plants, need to heat the water for the desalination process. Multi-effect desalination is a very efficient distillation process that can work at low temperatures between 70°C and 80°C (Bataineh, 2016). In this case, part of the heat from the geothermal fluid could be transferred to the water to be desalinated (represented by Equation 8), in which case, the thermal power could be used instead of converting the thermal power to electric power.

The resulting production rates for this case are also a function of the compaction curves for porosity and the permeability regression data used for static model generation. Additional data from the basin can provide more insights regarding porosity and permeability variations with depth which could further enhance the predictions and results from the simulations.

The simulations shows that for this a case where the compartment from which water is being produced is large enough as the one shown on Fig. 7 (a), pressure support could be high enough to produce water without the need for injectors resulting in a pressure drop of around 25%. However, this would strongly depend on reservoir size and connectivity with additional pore pressure. Reservoirs with more compartments due to faults can result in faster pressure drop from production and would require the injectors to maintain the production rate. Additionally, for those cases where the pore volume is lower, pore pressure could drop even further and would increase the risk of fault reactivation and induced seismicity.

Considering a possible poroelastic effect caused by pressure depletion in the basin, a reduction in pore pressure could result in a decrease in the horizontal stress which, depending on the present-day stress field in the area, could trigger fault reactivation due to shear failure in optimally oriented faults (Zoback, 2007). Because of this, water reinjection can be a solution for maintaining reservoir pressure while reducing risk of fault reactivation.

On Fig. 13 it was observed that although Producer 1 had a larger spacing from its injector compared to Producer 2, thermal breakthrough happened faster on Producer 1 which in this case was driven by higher permeability in the area near Producer 1. This shows that well spacing is not the only parameter influencing thermal breakthrough and heterogeneity will have a very important influence here that always needs to be considered. In reservoirs where facies control permeability anisotropy like in fluvial systems, faster thermal breakthrough can be expected if producers and injectors are placed parallel to channel orientation.

Conclusions

- Production rates of 80 liters/second with a temperature of around 120 °C could be achieved on the Al-Wajh Formation of the Yanbu Basin based on the reservoir parameters used here.
- Water reinjection provides pressure support for producers while maintaining reservoir pressure, avoiding significant changes in horizontal stress.
- A well spacing between 1400 m and 1750 m could provide sufficient pressure support while reducing the time for thermal breakthrough. This will be affected by local variations in permeability related to the depositional environment of the Formation that could reduce thermal breakthrough times.
- Placing the producers at greater depth increases the risk of finding the Formation with very low porosity and permeability drastically reducing its flow capacity.

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